

Probability & Random Processes EE2011

Department of Electrical Engineering

Revision of Concepts

**NATIONAL UNIVERSITY OF COMPUTER &
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Events

In Probability Theory subsets of the sample space are called *events*.

EXAMPLE : The set of basic outcomes of rolling a die *once* is

$$\mathcal{S} = \{ 1 , 2 , 3 , 4 , 5 , 6 \} ,$$

so the subset $E = \{ 2 , 4 , 6 \}$ is an example of an event.

If a die is rolled *once* and it lands with a 2 *or* a 4 *or* a 6 up then we say that the event E has *occurred*.

We have already seen that the probability that E occurs is

$$P(E) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} .$$

The Algebra of Events

Since events are *sets*, namely, subsets of the sample space $\mathcal{S}$, we can do the usual *set operations* :

If E and F are events then we can form

$$\begin{array}{ll} E^c & \text{the } \textit{complement} \text{ of } E \\ E \cup F & \text{the } \textit{union} \text{ of } E \text{ and } F \\ EF & \text{the } \textit{intersection} \text{ of } E \text{ and } F \end{array}$$

We write $E \subset F$ if E is a *subset* of F .

REMARK : In Probability Theory we use

$$E^c \quad \text{instead of} \quad \bar{E} ,$$

$$EF \quad \text{instead of} \quad E \cap F ,$$

$$E \subset F \quad \text{instead of} \quad E \subseteq F .$$

Axioms of Probability

A *probability function* P assigns a real number (the *probability* of E) to every event E in a sample space $\mathcal{S}$.

$P(\cdot)$ must satisfy the following basic properties :

- $0 \leq P(E) \leq 1$,
- $P(\mathcal{S}) = 1$,
- For any *disjoint events* E_i , $i = 1, 2, \dots, n$, we have
$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots P(E_n) .$$

EXAMPLE : Suppose a coin is tossed two times.

The sample space is

$$\mathcal{S} = \{HH, HT, TH, TT\} .$$

Let E be the event ”*two Heads*” , *i.e.*,

$$E = \{HH\} .$$

Let F be the event ”*the first toss gives Heads*” , *i.e.*,

$$F = \{HH, HT\} .$$

Then

$$EF = \{HH\} = E \quad (\text{ since } E \subset F) .$$

We have

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{P(E)}{P(F)} = \frac{\frac{1}{4}}{\frac{2}{4}} = \frac{1}{2} .$$

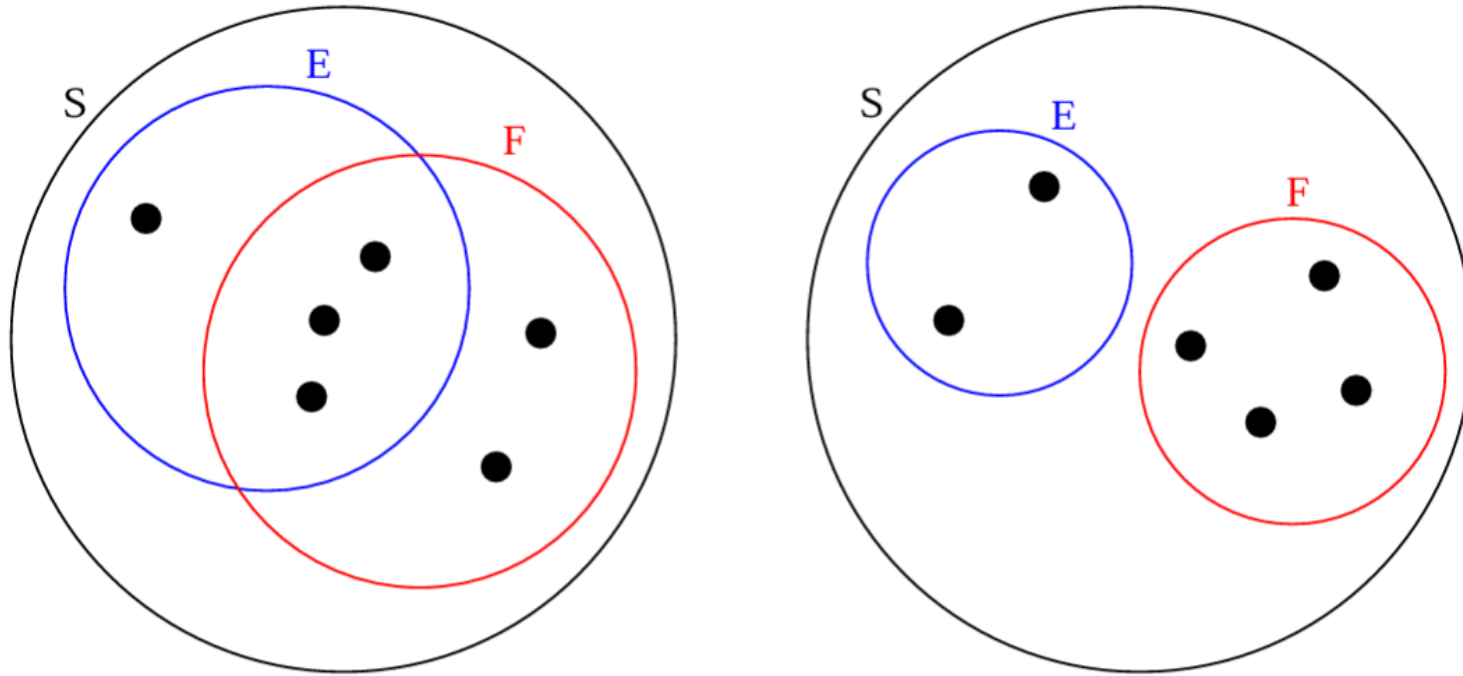
Suppose we draw a card from a shuffled set of 52 playing cards.

- What is the probability of drawing a Queen ?
- What is the probability of drawing a Queen, given that the card drawn is of *suit* Hearts ?
- What is the probability of drawing a Queen, given that the card drawn is a *Face card* ?

What do the answers tell us?

(We'll soon learn the events "Queen" and "Hearts" are *independent*.)

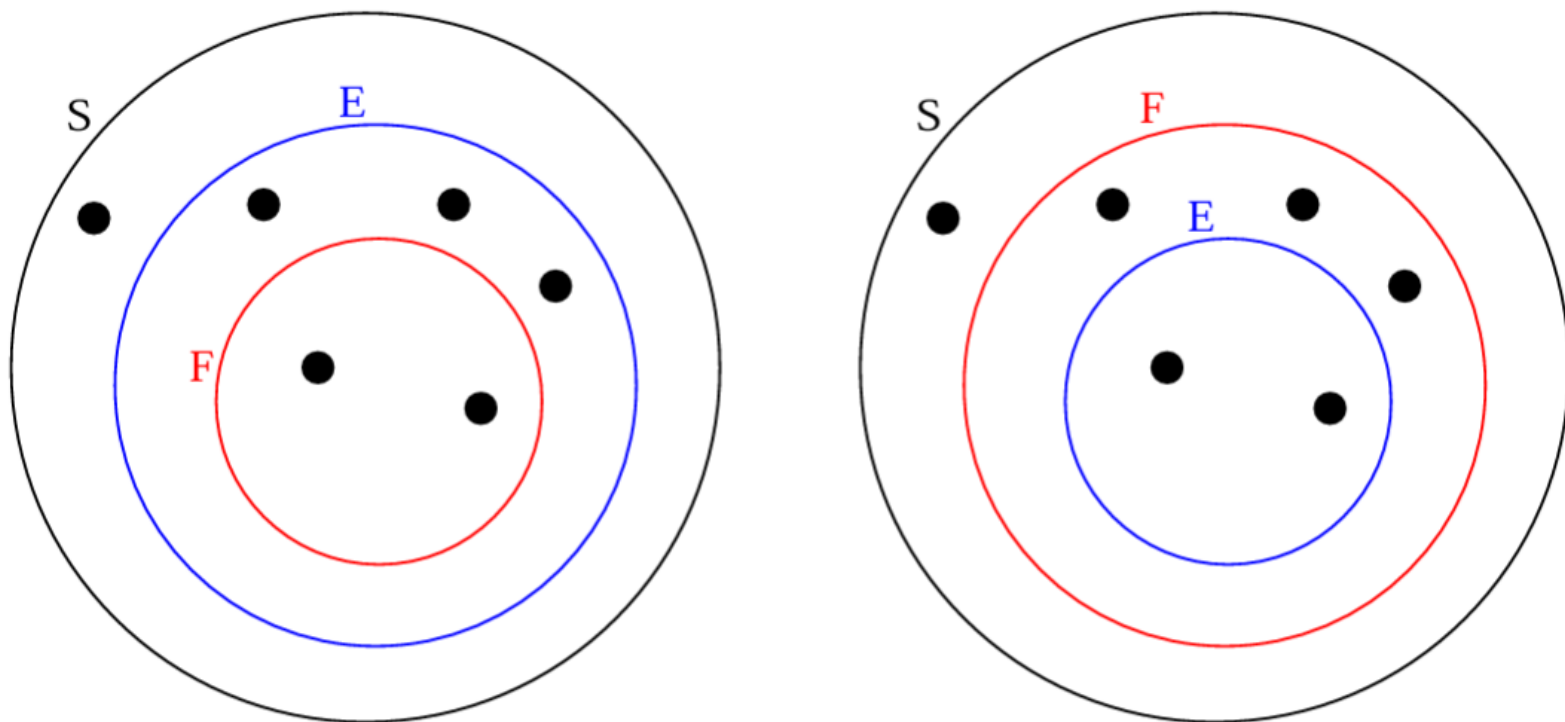
$$P(E|F) \equiv \frac{P(EF)}{P(F)}$$



Suppose that the 6 outcomes in $\mathcal{S}$ are equally likely.

What is $P(E|F)$ in each of these two cases ?

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EXAMPLE :

Suppose we draw a card from a shuffled set of 52 playing cards.

- What is the probability of drawing a Queen, given that the card drawn is of *suit* Hearts ?

ANSWER :

$$P(Q|H) = \frac{P(QH)}{P(H)} = \frac{\frac{1}{52}}{\frac{13}{52}} = \frac{1}{13} .$$

What is the probability of drawing a Queen, given that the card drawn is a *Face card* ?

ANSWER :

$$P(Q|F) = \frac{P(QF)}{P(F)} = \frac{P(Q)}{P(F)} = \frac{\frac{4}{52}}{\frac{12}{52}} = \frac{1}{3} .$$

(Here $Q \subset F$, so that $QF = Q$.)

Example:

A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75.

Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers.

A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Solution:

Consider the events:

A : a customer purchases latex paint,

A' : a customer purchases semigloss paint,

B : a customer purchases rollers.

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A')P(A')} = \frac{(0.60)(0.75)}{(0.60)(0.75) + (0.25)(0.30)} = 0.857.$$